

Problem:- Predict house price with the help of area.

$X \rightarrow Y$
 $\overline{g} \quad \overline{D}$

↑ always 1 in number

↑ (≥ 1 in number)

✓ Target variable
or

✓ independent variable

✓ Dependent variable
or

✓ X-variable
or

✓ y-variable

✓ features/attributes

Area $\rightarrow$ Price

Area, no. of floors,
age of house etc $\rightarrow$ Price ✓

Q:- Price = 35L
33 m² =

Relationship
 exists
 we are not
 able to find

No

Sol

Approximate

future
 relationship

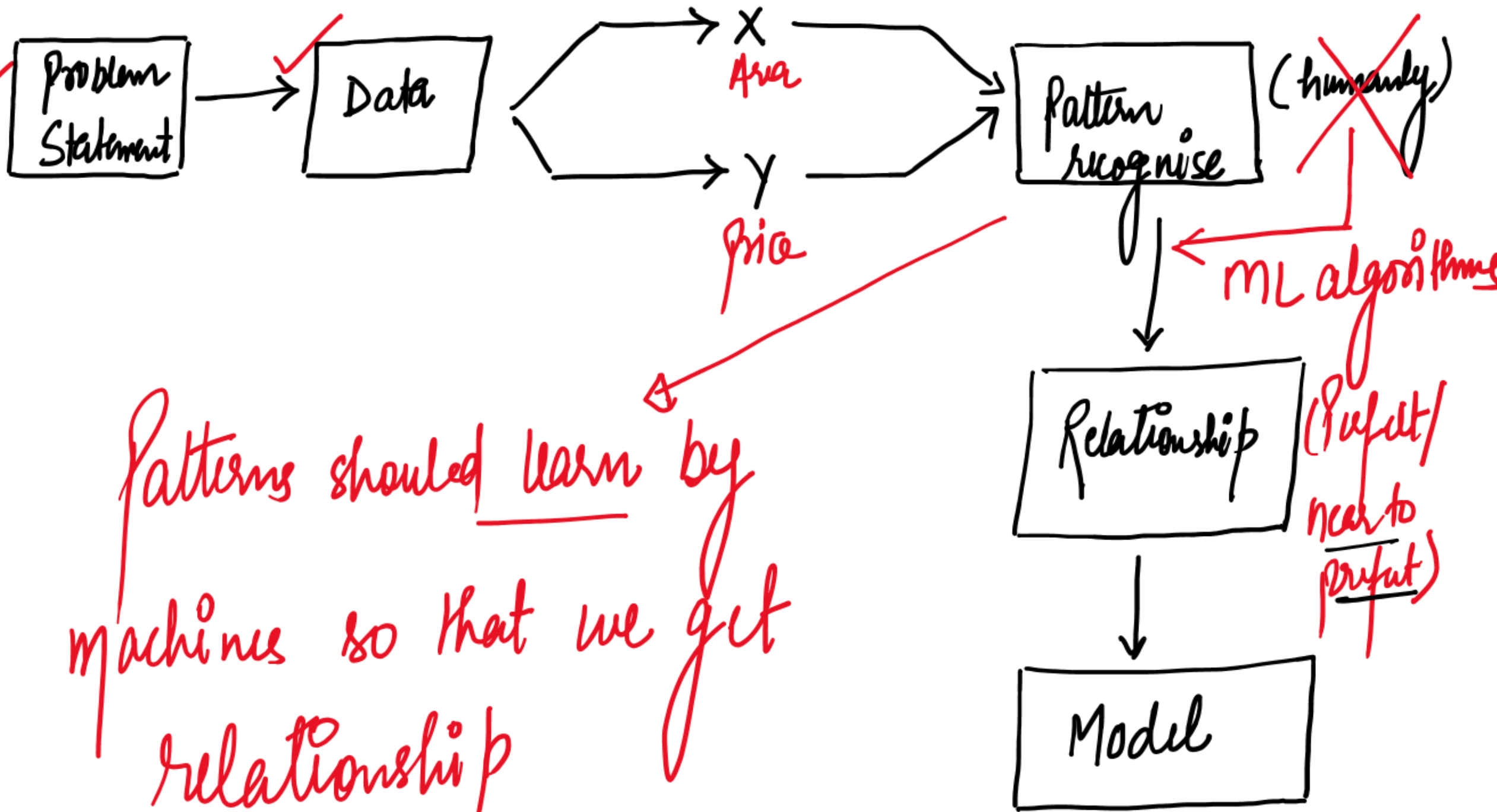
particle

SLR

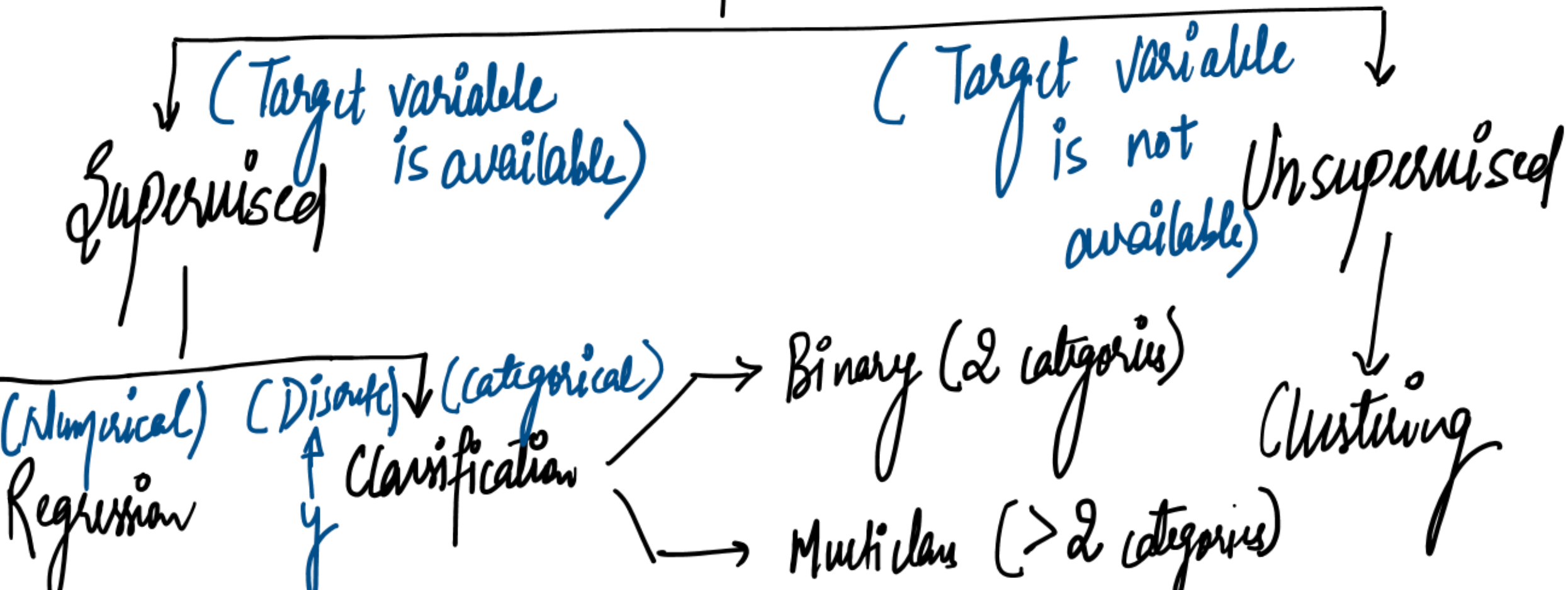
~~$y = x + 5$~~

34.5

X (m ²)		Y
Area		Price (L)
30		32
50		48
42		46
24		20



ML Algorithms

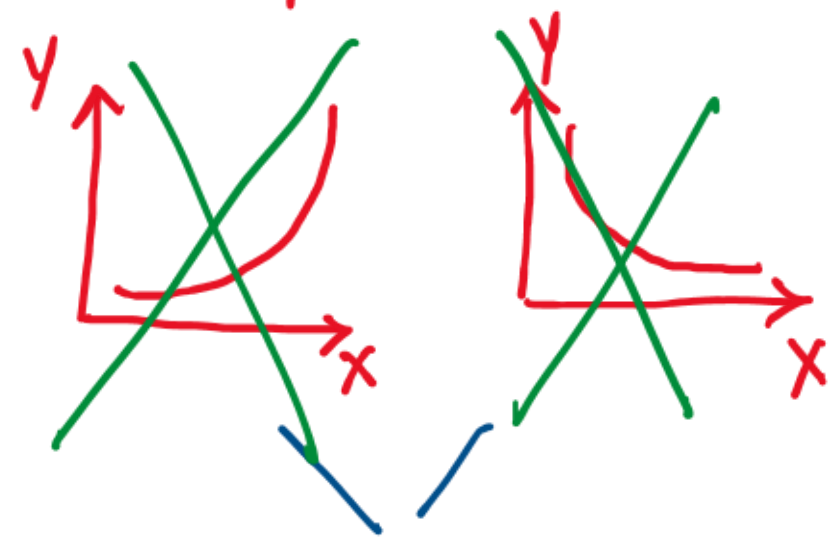


→ Linear Regression:

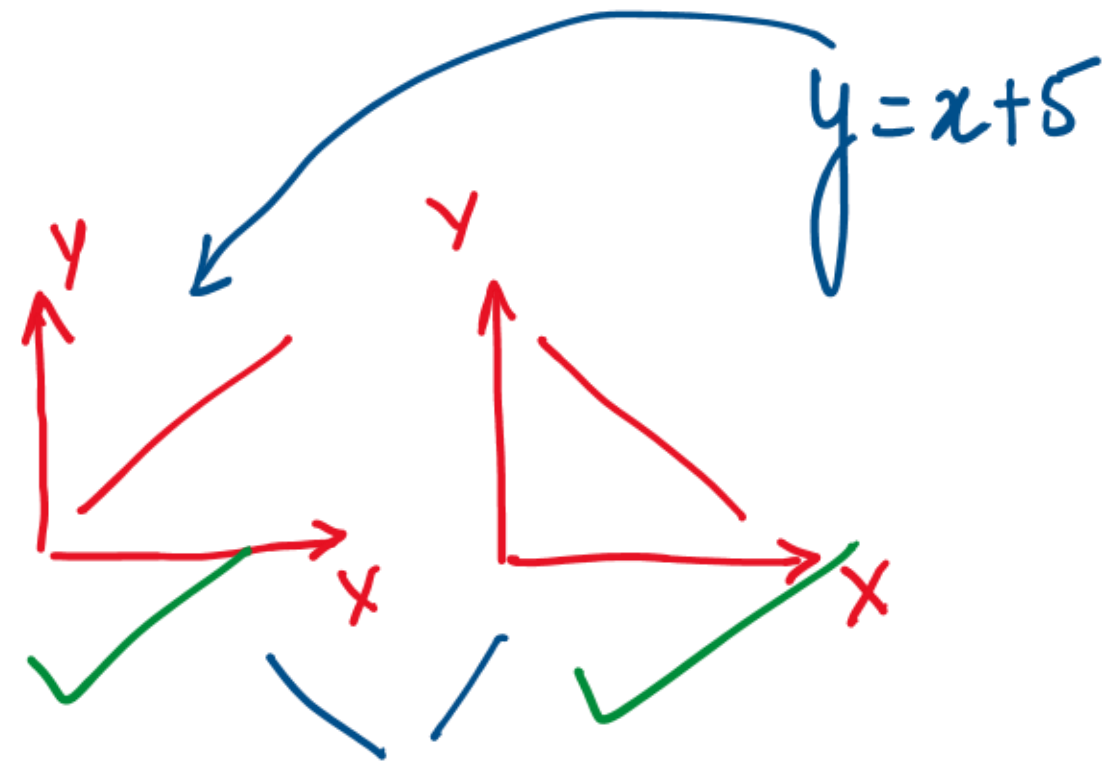
predicts numerical column

$X \rightarrow Y$

(Linear relationship)

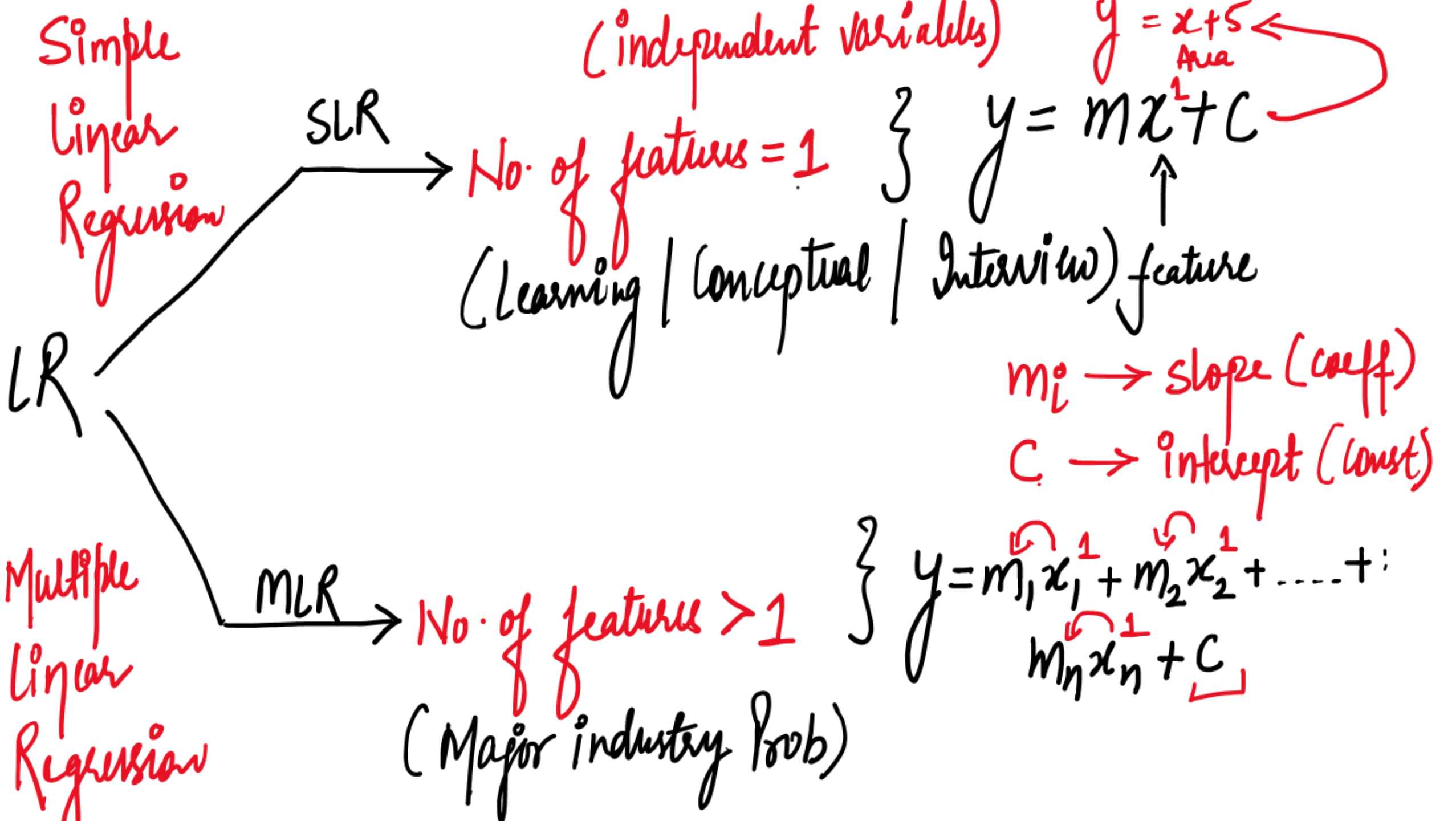


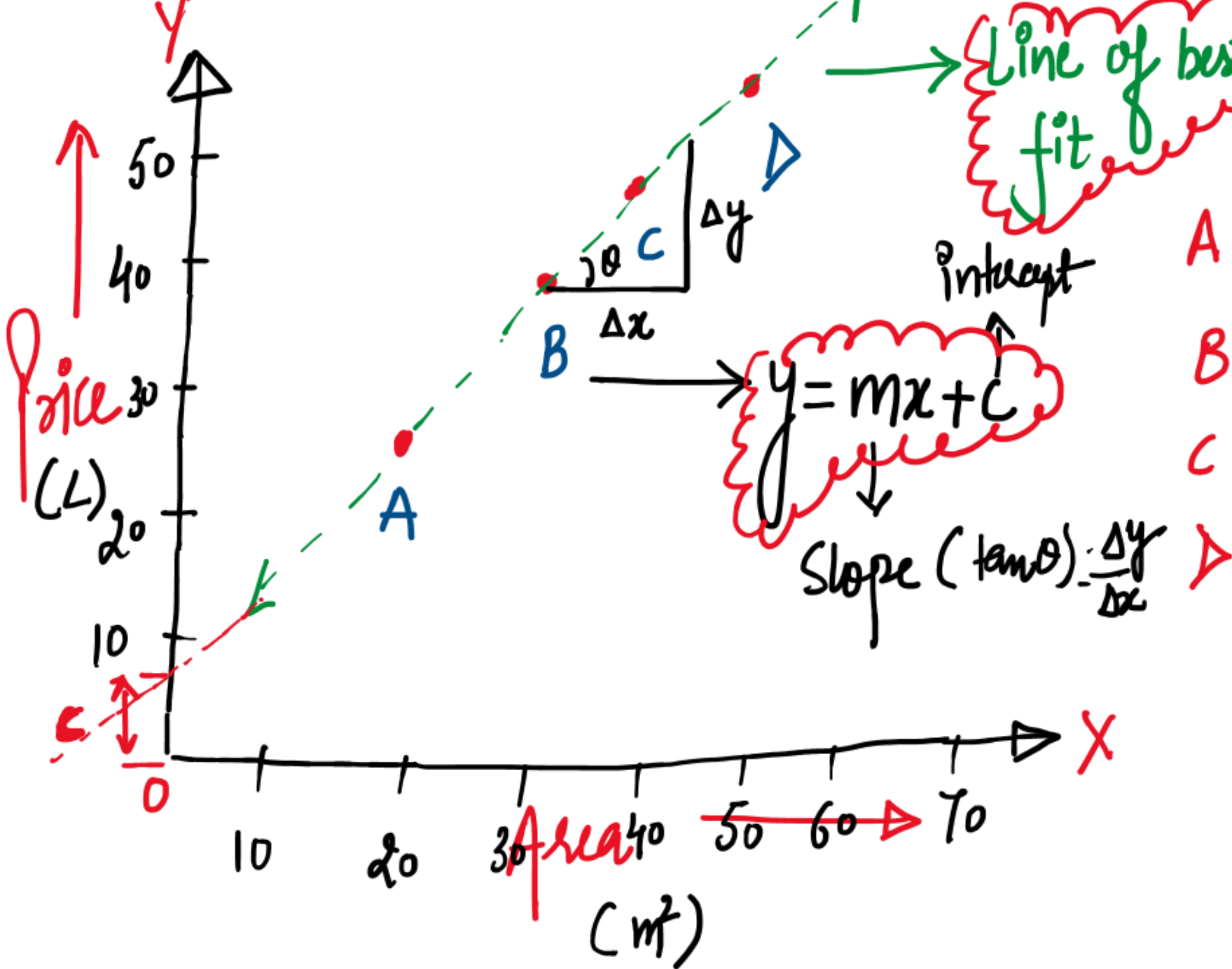
Non-linear



$$y = x + 5$$

Linear

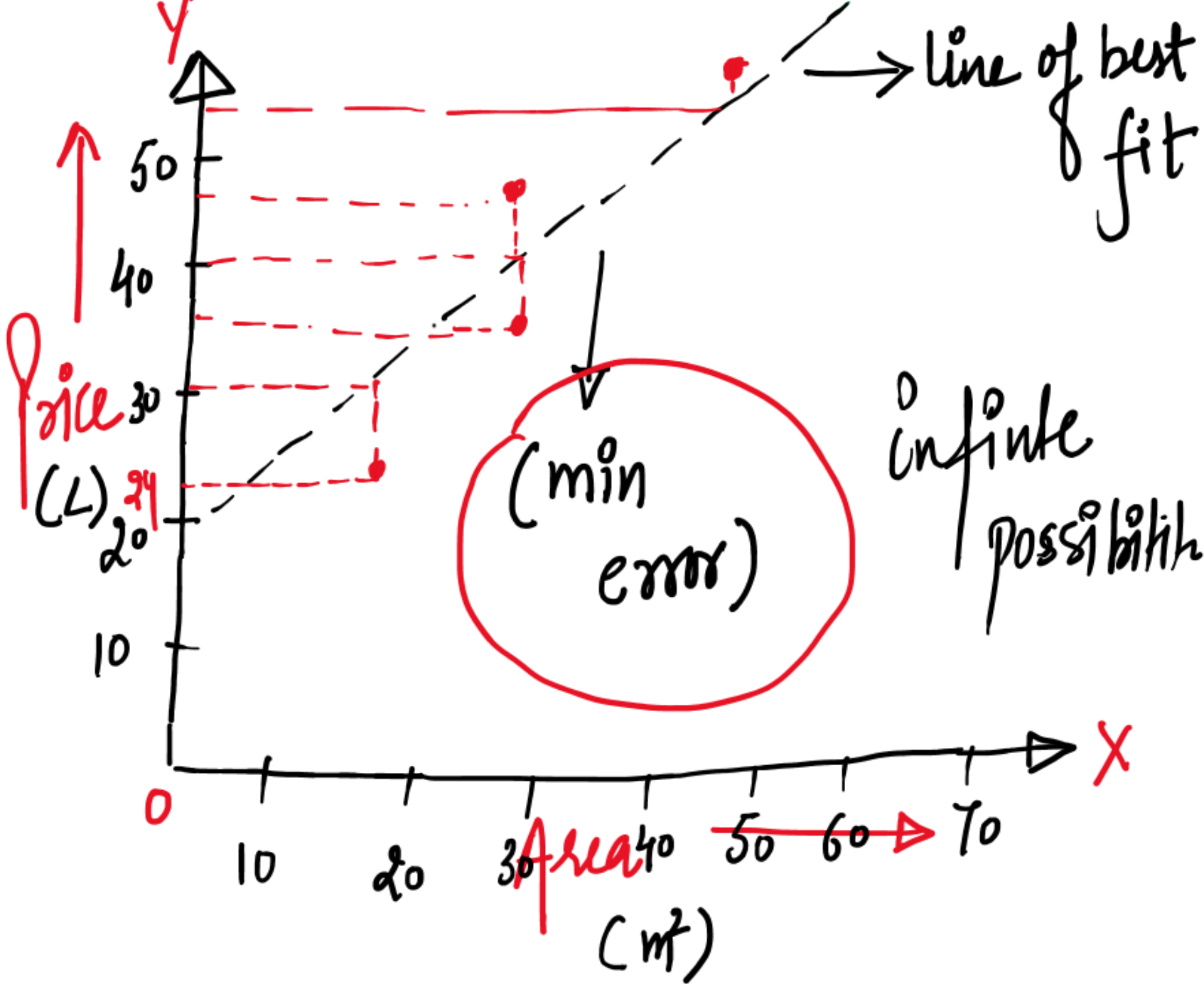




Area	Price
X	Y
<u>20</u>	25
<u>32</u>	37
38	43
50	55

$$\left. \begin{aligned} y &= \beta_0 + \beta_1 x \\ y &= \alpha_0 + \alpha_1 x \end{aligned} \right\}$$

$\beta_0 \leftarrow c$ $\beta_1 \leftarrow m_1$



Area	Price	Price (y _p)
X	Y (A _p)	
20	25 24	30 = 30
32	37 38	40 = 40
38	43 42	40 = 40
50	55 56	55 = 55

$$y_a = \underbrace{mx + c}_{y_p} + \underbrace{\epsilon}_{\text{Error term}}$$

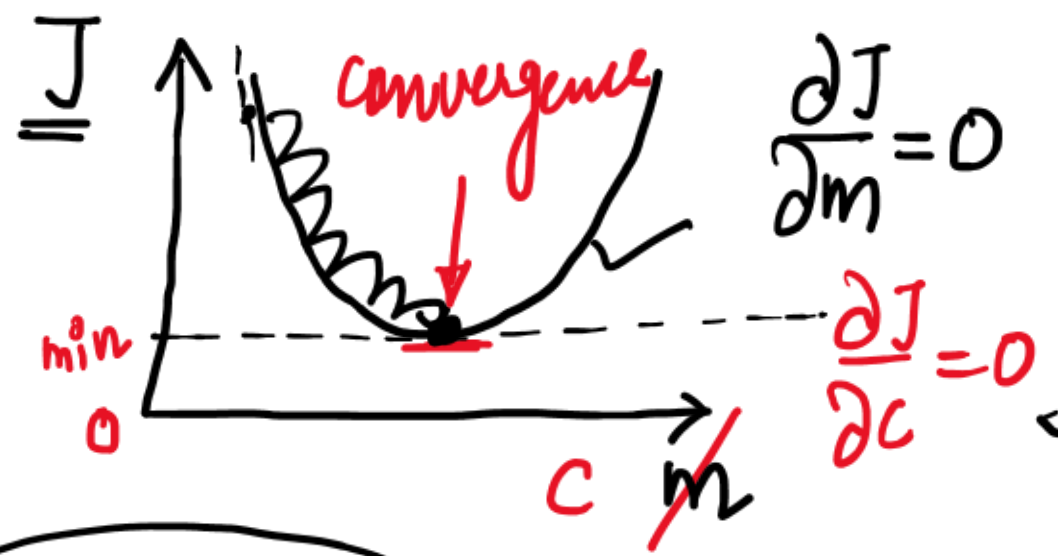
$\rightarrow \underline{\text{Error } f^n} / \underline{\text{Cost } f^n} / \underline{\text{Loss } f^n} :- (\underline{\text{Mean Square Error}}) \frac{\text{Sum}}{N} \text{ (MSE)}$

$$\left\{ J = \frac{1}{N} \sum_{i=1}^N [y_i - (mx + c)]^2 \right\}$$

line $\rightarrow$ minimum $\rightarrow$ line of best fit

→ Gradient Descent Algorithm:-

① Initialise m & C as zero.



② Update the values of m & C

fit()

$$m' = m - \alpha \frac{\partial J}{\partial m}$$

$$C' = C - \alpha \frac{\partial J}{\partial C}$$

{ Error gradient }

$\alpha \rightarrow$ Learning rate = 0.01

③ Repeat ② again & again until convergence. (min Error)

→ Evaluation Metrics:-

$$TSS = \underbrace{CSS}_{\uparrow} + \underbrace{RSS}_{\uparrow x}$$

*

$$R^2 = \frac{\checkmark \text{ Captured Squared Error (CSS)}}{\checkmark \text{ Total Sum of Squared Errors (TSS)}} \\ = \frac{TSS - \text{Residual Squared Error (RSS)}}{TSS}$$



$$R^2 = 1 - \frac{RSS}{TSS}$$

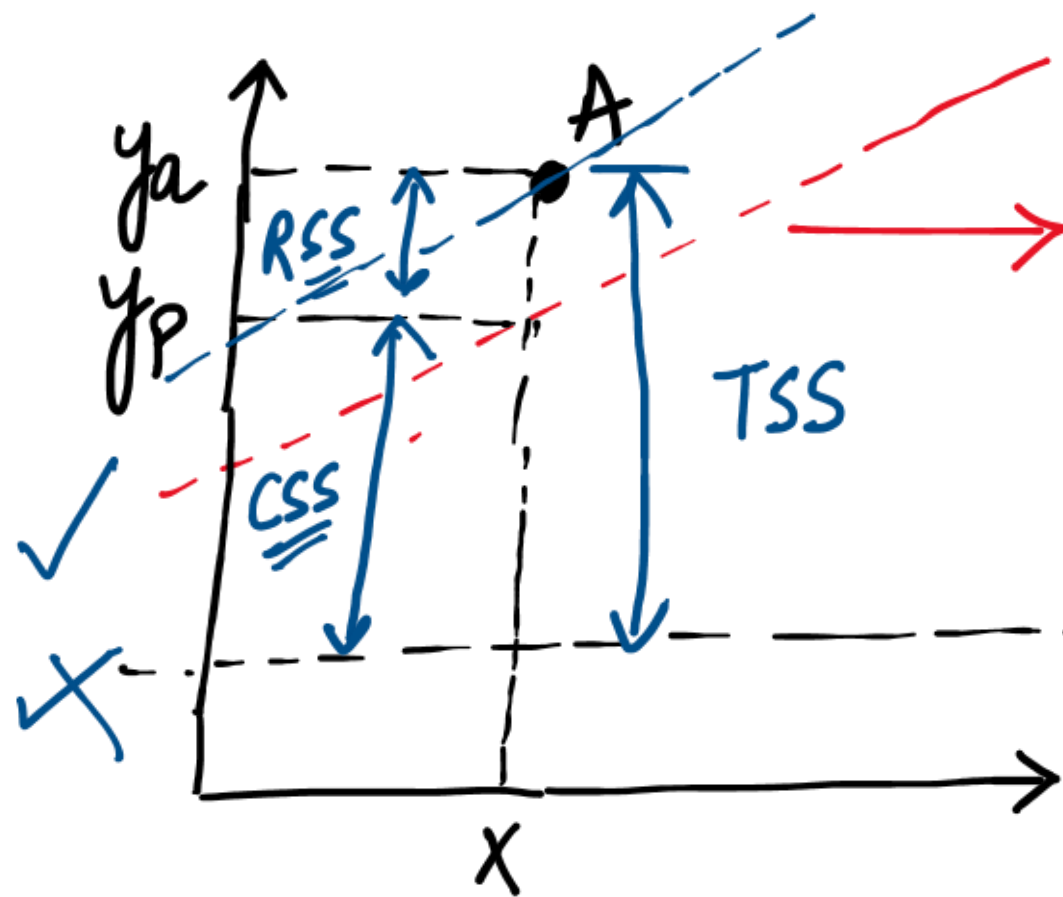
$$R^2 \in [0, 1]$$

$$TSS = CSS + RSS$$

$$\max \{ RSS = TSS \} \quad \min \{ RSS = 0 \}$$

> 70% or 80%

Industry benchmark



$$y = mx + c$$

$$\bar{y} = \frac{\sum y}{N} = \text{Average model}$$

Reference model

$$R^2 = 0$$

$$TSS = RSS$$

